Homework 3: Sequence-to-Sequence Machine Translation

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Assignment: Homework 3

GitHub Repository: https://github.com/gfeliu-rgb/ECGR-5106-HW3

Dataset and Experimental Setup

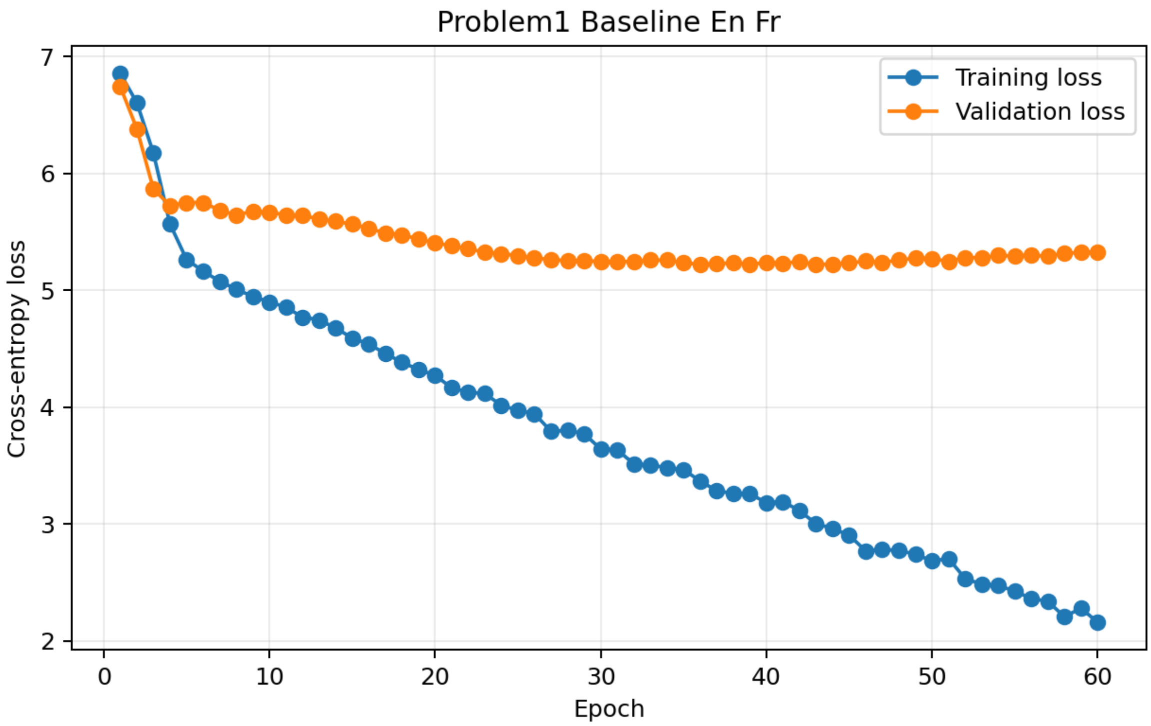
The provided vast_english_french.txt dataset contains 555 English/French sentence pairs. I used one deterministic 80/20 split with seed 4106, producing 444 training examples and 111 validation examples. The same split was reused across all English-to-French and French-to-English experiments.

Text was normalized by lowercasing, removing accent marks, separating punctuation, and collapsing whitespace. Each vocabulary used , , , and tokens. Models were trained with token-level cross-entropy loss, Adam optimization, gradient clipping, teacher forcing during training, and greedy decoding during validation.

Model	Best Val Loss	Exact Acc.	Word BLEU-4	Char BLEU-4	Char Acc.	Seq. Sim.
Baseline EN-FR	5.2227	0.0000	0.0276	0.1463	0.1483	0.4289
Attention EN-FR	5.0146	0.0000	0.0589	0.2431	0.1835	0.4688
Baseline FR-EN	4.8302	0.0000	0.0249	0.1414	0.1675	0.4229
Attention FR-EN	4.6855	0.0000	0.0771	0.2481	0.1955	0.4697

Problem 1: Baseline GRU Encoder-Decoder

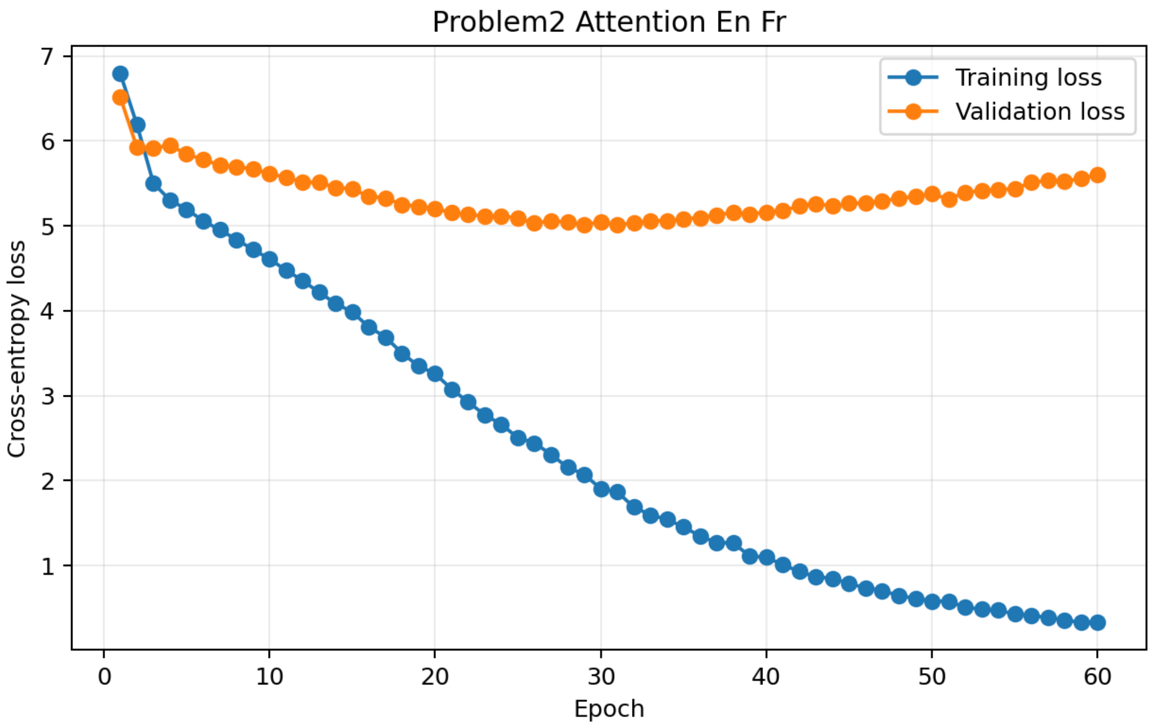
The baseline English-to-French model used a GRU encoder and GRU decoder. The final encoder hidden state initialized the decoder, which generated French tokens one at a time. The model used source vocabulary size 901, target vocabulary size 996, embedding size 128, hidden size 160, batch size 96, and 60 epochs.

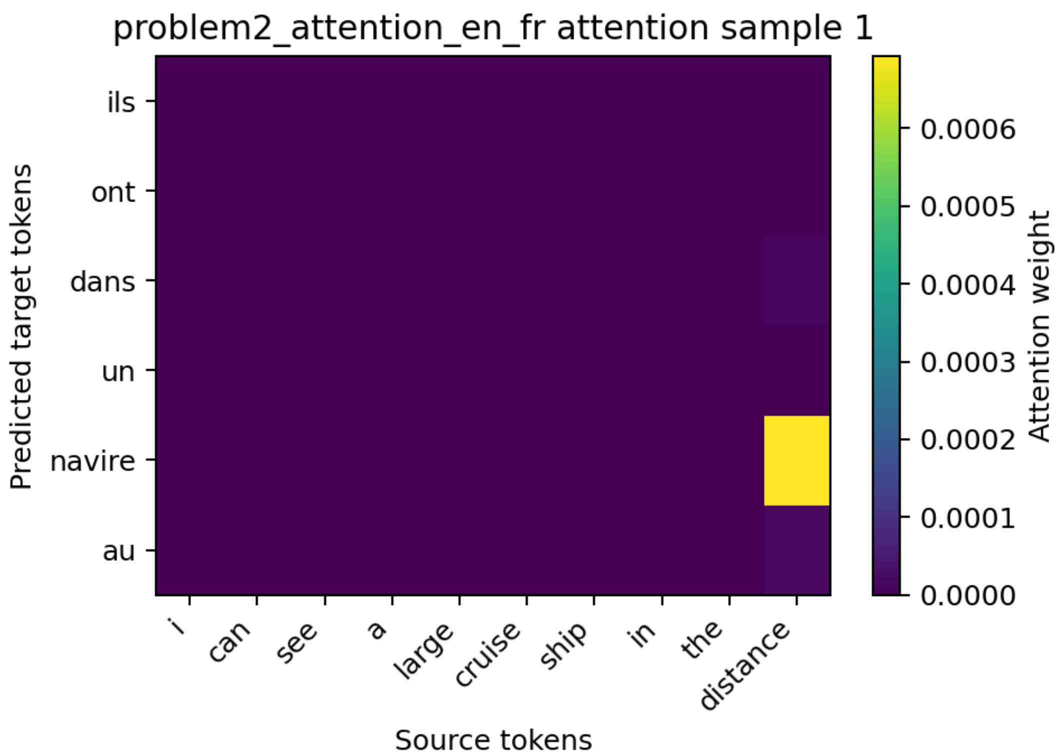


The best validation loss was 5.2227. Traditional exact sequence accuracy was 0.0000 and validation word BLEU-4 was 0.0276. The nonzero BLEU and sequence-similarity values show partial word overlap, but the model did not generate complete exact sentence matches.

Problem 2: GRU Encoder-Decoder with Luong Attention

The attention model used the same GRU backbone but added Luong dot-product attention. At each decoding step, the decoder hidden state scored all encoder outputs to form a context vector, giving the decoder direct access to source-token information.



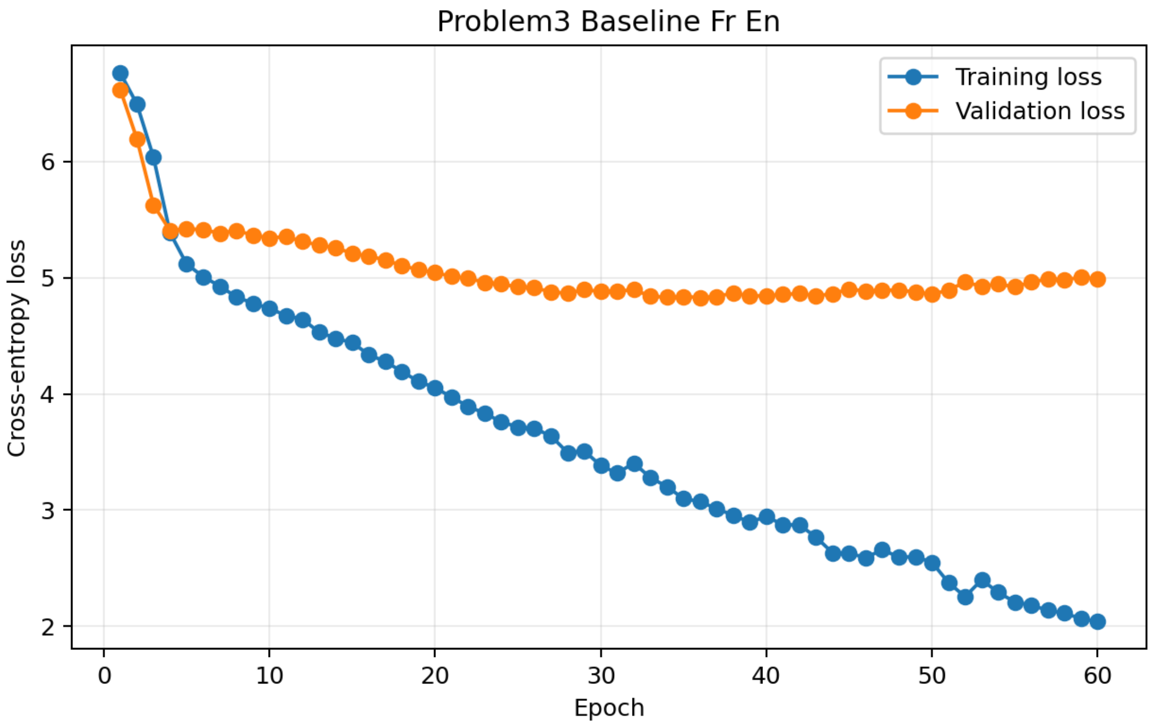


Attention improved English-to-French word BLEU-4 from 0.0276 to 0.0589 and improved best validation loss from 5.2227 to 5.0146. Exact-match accuracy remained 0.0000 because exact match requires every generated word to match the reference perfectly, while BLEU-4 gives partial credit for overlapping n-grams.

Problem 3: Reversing the Language Direction

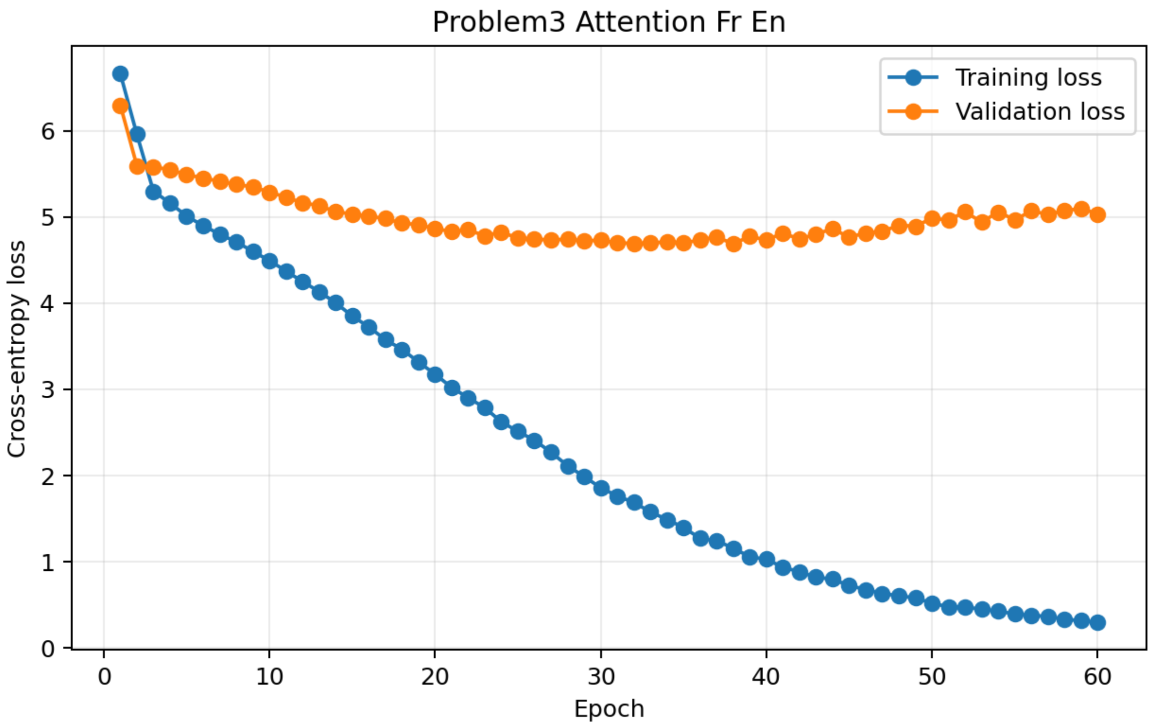
For French-to-English translation, I reused the same split and reversed each pair so French became the source language and English became the target language. I trained both the baseline GRU and the attention GRU under the same general configuration.

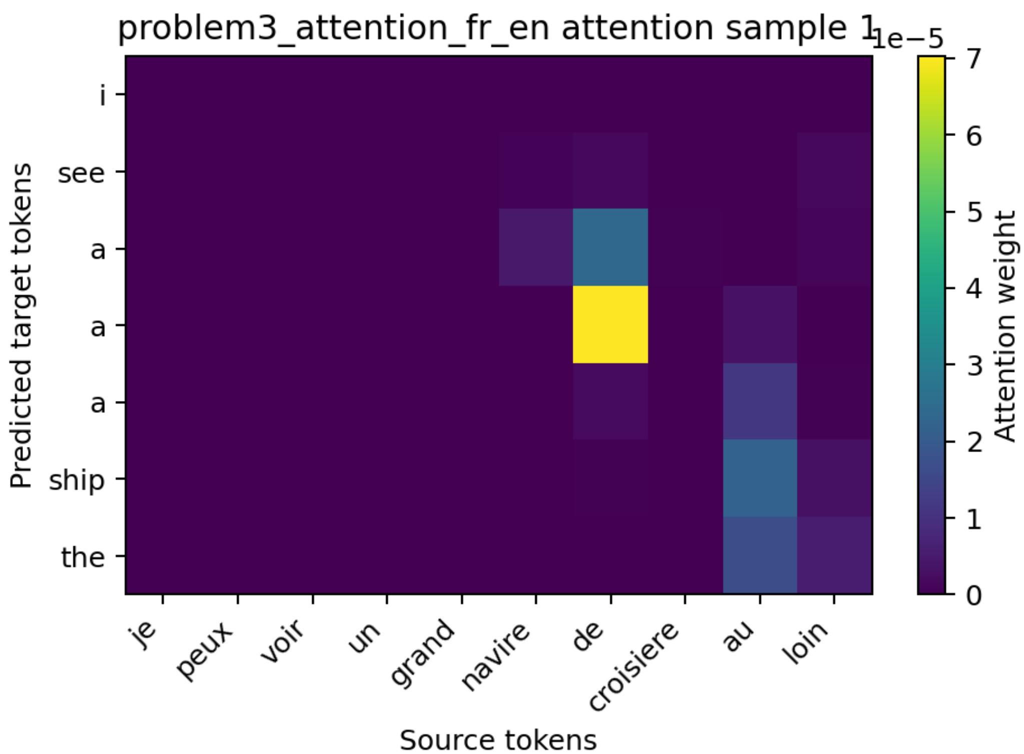
Baseline French-to-English



The reversed baseline model reached best validation loss 4.8302, exact sequence accuracy 0.0000, and word BLEU-4 0.0249.

Attention French-to-English





The reversed attention model reached best validation loss 4.6855, exact sequence accuracy 0.0000, and word BLEU-4 0.0771. This was the strongest result across the four experiments.

French-to-English appeared easier for the attention model to optimize because it achieved the best validation loss and highest BLEU-4. Attention improved both directions, but exact sequence accuracy stayed low because the validation metric is strict at the whole-sentence level.

Conclusion

The baseline GRU established a working sequence-to-sequence translation pipeline. Adding Luong attention improved validation loss, word BLEU-4, character BLEU-4, character accuracy, and sequence similarity in both translation directions. The French-to-English attention model performed best overall under the selected hyperparameters.